

Quantum Computation

Module 2 Assignment

Single-Qubit State Vectors, Pauli Gates and the Hadamard Gate in Qiskit

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Question 1.

The single qubit state vector $|\psi\rangle = a_0|0\rangle + a_1|1\rangle$

Let show that the following ket is valid state vector by showing its magnitude is normalized to

$$|\psi\rangle = \frac{i}{2}|0\rangle + \frac{\sqrt{3}e^{i\pi/6}}{2}|1\rangle$$

- step 1 - Let identify the amplitudes:

For $|\psi\rangle = a_0|0\rangle + a_1|1\rangle$, the amplitudes:

$$a_0 = \frac{i}{2}, \quad a_1 = \frac{\sqrt{3}e^{i\pi/6}}{2}$$

- step 2 - Squared magnitude of a_0

Let multiply the amplitude by its complex conjugate:

$$|a_0|^2 = a_0^* a_0 = \left(\frac{i}{2}\right)\left(\frac{-i}{2}\right) = \frac{-i^2}{4} = \frac{-(-1)}{4} = \frac{1}{4}$$

- step 3 - Squared magnitude of a_1 : The phase has magnitude one,

$$\text{so: } |a_1|^2 = \left(\frac{\sqrt{3}e^{i\pi/6}}{2}\right)\left(\frac{\sqrt{3}e^{-i\pi/6}}{2}\right) = \frac{3}{4}e^{i\pi/6 - i\pi/6} = \frac{3}{4}e^0 = \frac{3}{4}$$

step 4 - Normalization

$$\langle\psi|\psi\rangle = |a_0|^2 + |a_1|^2 = \frac{1}{4} + \frac{3}{4} = 1$$

Thus the norm is the square root of 1, which is 1. The given ket is a valid normalized state vector

Question 2

Implement (a) $H|0\rangle$, (b) $Y|0\rangle$, and (c) $Z|0\rangle$ using `qc = QuantumCircuit(1, 1)`. Each circuit below follows the assignment order:

- (i) Measure the output and show the histogram screenshot.
- (ii) Show the circuit screenshot produced by `draw()`.
- (iii) Derive the output state and $P(|0\rangle)$, $P(|1\rangle)$, then compare with the histogram.
- (iv) Collect all answers in this report, alongside the supplied Question 1 answer.

The computational basis is $|0\rangle = (1, 0)^T$ and $|1\rangle = (0, 1)^T$. X , Y , and Z are the Pauli operators; H is the Hadamard gate. The imaginary unit satisfies $i^2 = -1$.

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Gate actions and Qiskit syntax

Gate	Qiskit call	Action on $ 0\rangle$	Action on $ 1\rangle$	Intuition
X	<code>qc.x(0)</code>	$ 1\rangle$	$ 0\rangle$	Bit flip (quantum NOT)
Y	<code>qc.y(0)</code>	$i 1\rangle$	$-i 0\rangle$	Bit flip + phase flip
Z	<code>qc.z(0)</code>	$ 0\rangle$	$- 1\rangle$	Phase flip
H	<code>qc.h(0)</code>	$\frac{ 0\rangle + 1\rangle}{\sqrt{2}}$	$\frac{ 0\rangle - 1\rangle}{\sqrt{2}}$	Creates equal superposition

Qiskit implementation

Each circuit starts in $|0\rangle$ and has one classical bit. The output state vector is computed before measurement. Measurement in the computational basis stores the result in classical bit 0. All reported runs use an ideal AerSimulator with 1024 shots and seed 42.

For $b_0|0\rangle + b_1|1\rangle$, the Born rule gives $P(|0\rangle) = |b_0|^2$ and $P(|1\rangle) = |b_1|^2$. Observed frequencies are counts divided by 1024.

```
import numpy as np
from IPython.display import display
from qiskit import QuantumCircuit, transpile
from qiskit.quantum_info import Statevector
from qiskit.visualization import plot_histogram
from qiskit_aer import AerSimulator

def run_circuit(gate):
    qc = QuantumCircuit(1, 1)
    getattr(qc, gate)(0)
    state = Statevector.from_instruction(qc)
    qc.measure(0, 0)
    sim = AerSimulator()
    compiled = transpile(qc, sim, seed_transpiler=42)
    raw = sim.run(compiled, shots=1024,
                  seed_simulator=42).result().get_counts()
    counts = {k: raw.get(k, 0) for k in ("0", "1")}
    print("Statevector:", np.round(state.data, 4))
    print("Counts:", counts)
    print("Observed frequencies:",
          {k: v / 1024 for k, v in counts.items()})
    display(plot_histogram(counts))
    display(qc.draw("mpl"))
    return qc, state, counts
```

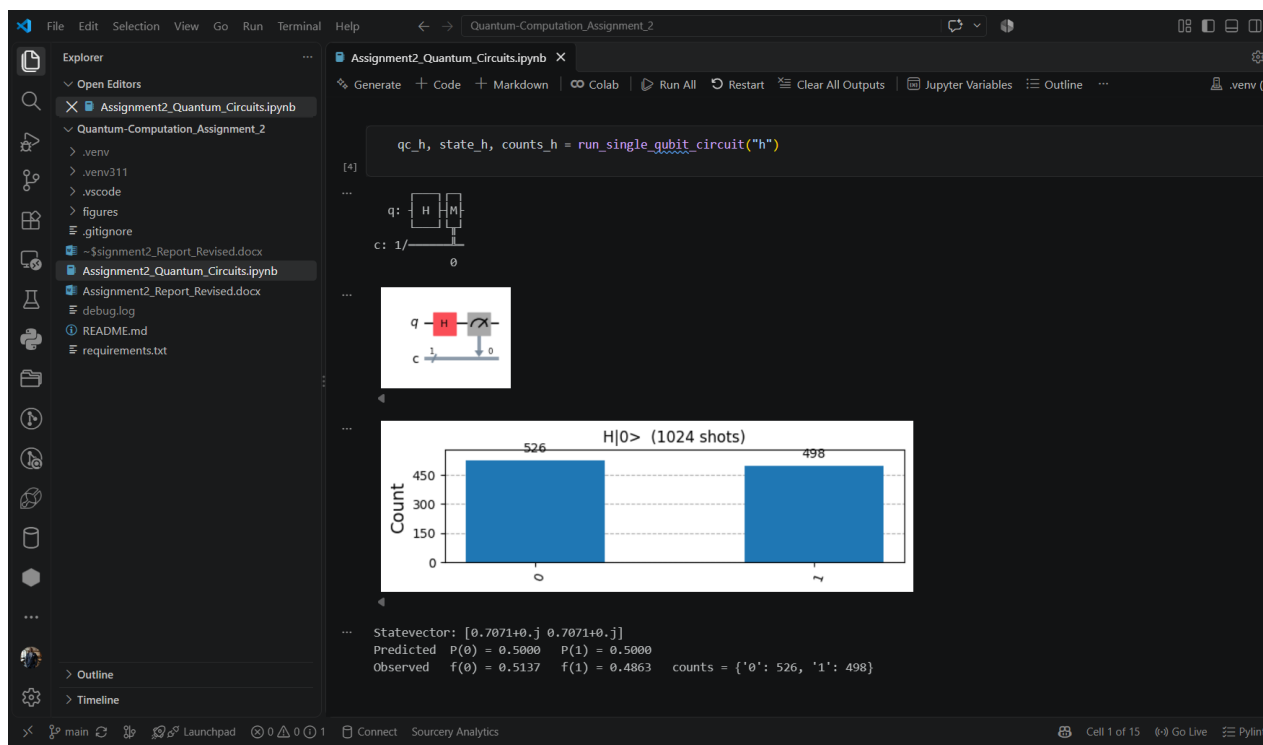
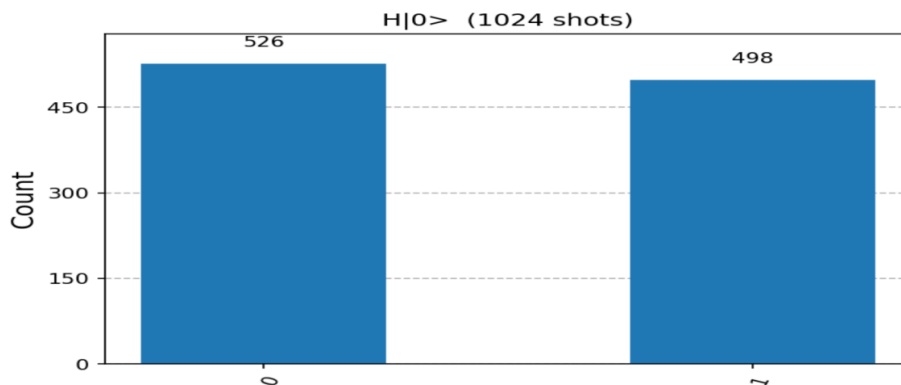
(a) $H|0\rangle$ — Hadamard gate applied to $|0\rangle$

(i) Measurement histogram for $H|0\rangle$

Apply `qc.h(0)` to `QuantumCircuit(1, 1)`. Run `qc_h, state_h, counts_h = run_circuit("h")` using the shared setup. Measure with `qc.measure(0, 0)`.

1024-shot output: `{'0': 526, '1': 498}`

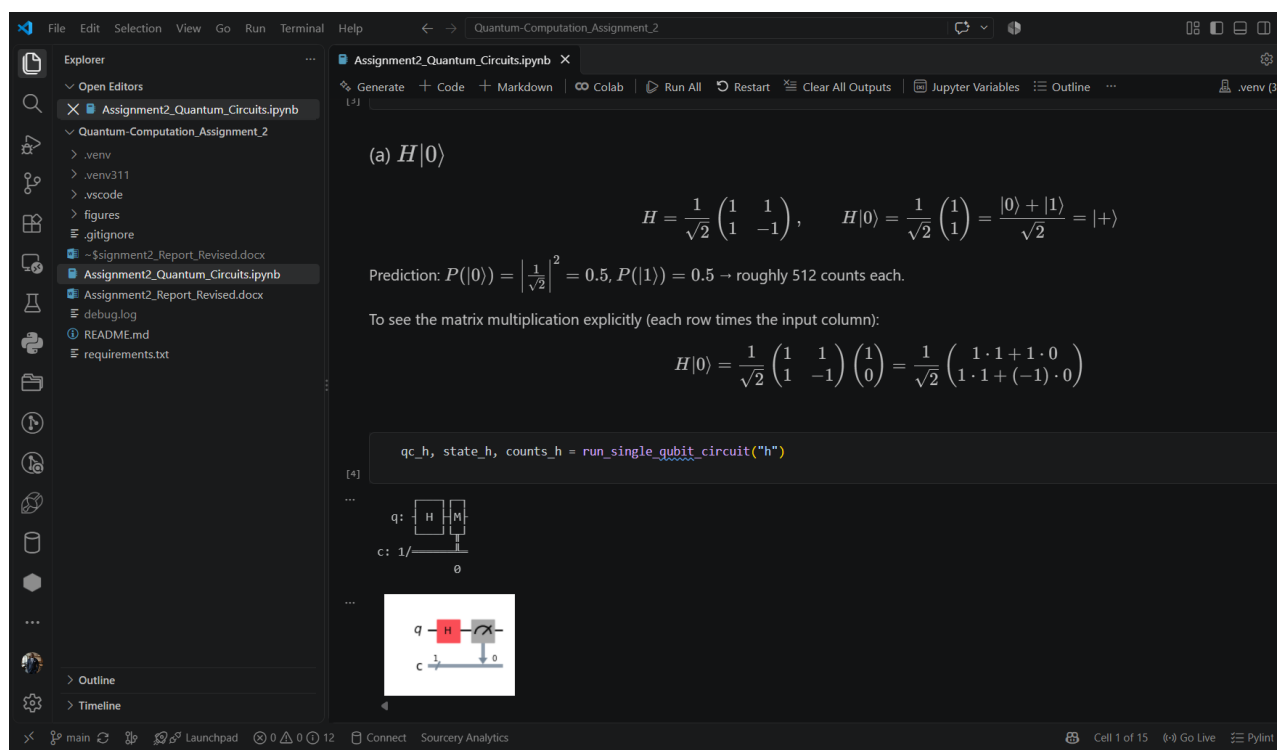
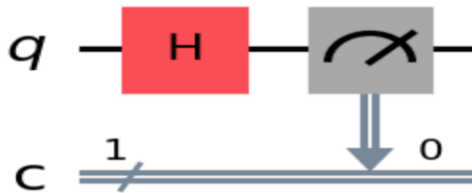
State vector before measurement: `[0.7071+0.j, 0.7071+0.j]`



$H|0\rangle$: measurement histogram and executed-notebook screenshot.

(ii) Circuit screenshot for $H|0\rangle$ after running draw()

The measured circuit is drawn with `qc.draw("mpl")`. It contains one H gate on qubit 0 followed by measurement into classical bit 0.



$H|0\rangle$: circuit diagram and executed-notebook screenshot.

(iii) $H|0\rangle$ - Hadamard gate applied to $|0\rangle$
 Derive the output state and probabilities, verify the histogram.

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \times 1 + 1 \times 0 \\ 1 \times 1 + (-1) \times 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2} = |+\rangle$$

Predicted measurement probabilities for $H|0\rangle$

$$P(|0\rangle) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}, \quad P(|1\rangle) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

The probabilities sum to 1. For 1024 shots, the expected counts are 512 zeros and 512 ones. The state vector agrees with Qiskit's $[0.7071 + 0.j, 0.7071 + 0.j]$.

The measures

- Verification against the $H|0\rangle$ histogram.
 the measured frequencies are $526/1024 = 0.513671875$ and $498/1024 = 0.486328125$, both close to the predicted 0.5. Each differs by 0.013671875.

The standard deviation of the zero count is

$\sqrt{1024 \times 0.5 \times 0.5} = 16$
 The difference from 512 is 14 or 0.875 standard deviation
 This is consistent with finite sampling; equal probabilities do not require exactly equal counts in a single run.

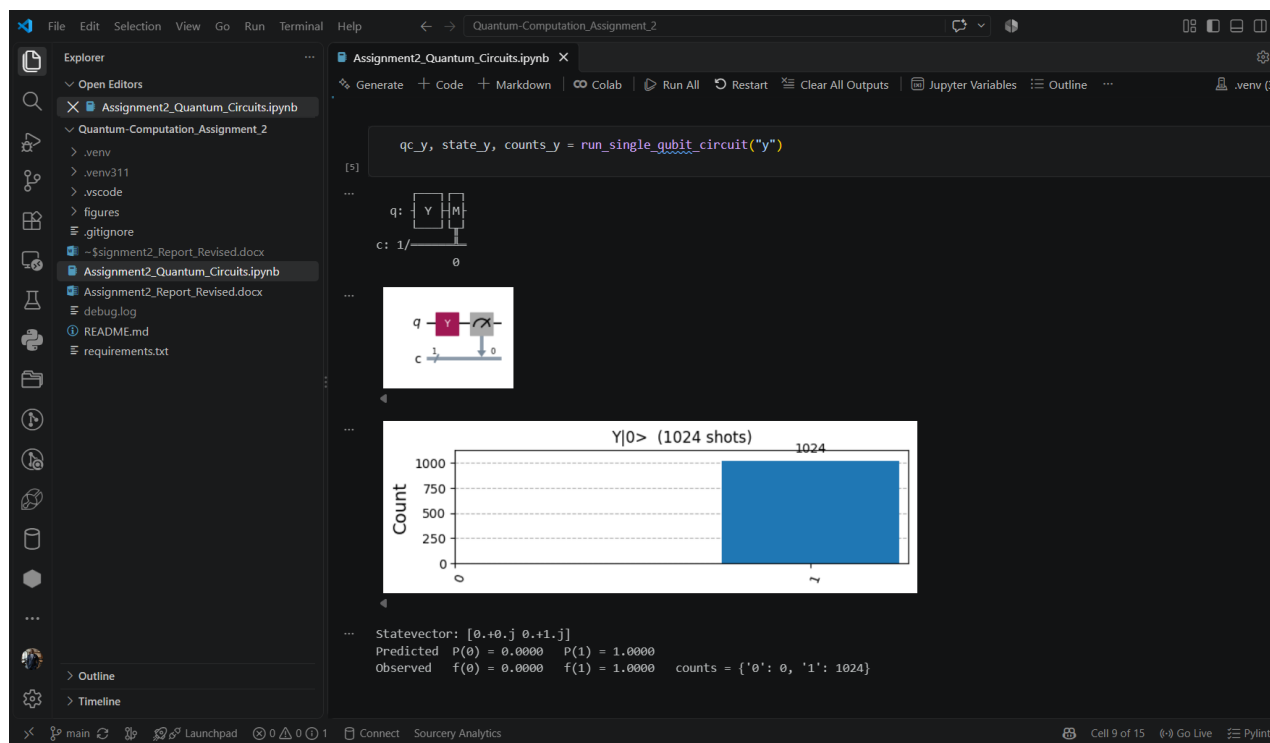
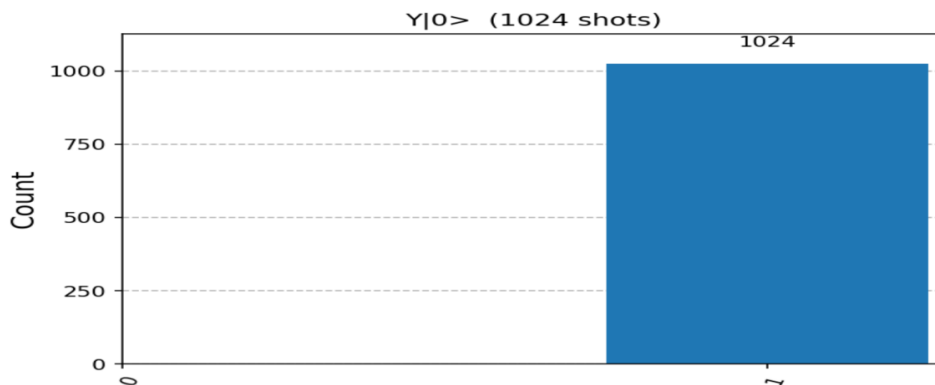
(b) $Y|0\rangle$ — Pauli Y gate applied to $|0\rangle$

(i) Measurement histogram for $Y|0\rangle$

Apply `qc.y(0)` to `QuantumCircuit(1, 1)`. Run `qc_y, state_y, counts_y = run_circuit("y")` using the shared setup. Measure with `qc.measure(0, 0)`.

1024-shot output: `{'0': 0, '1': 1024}`

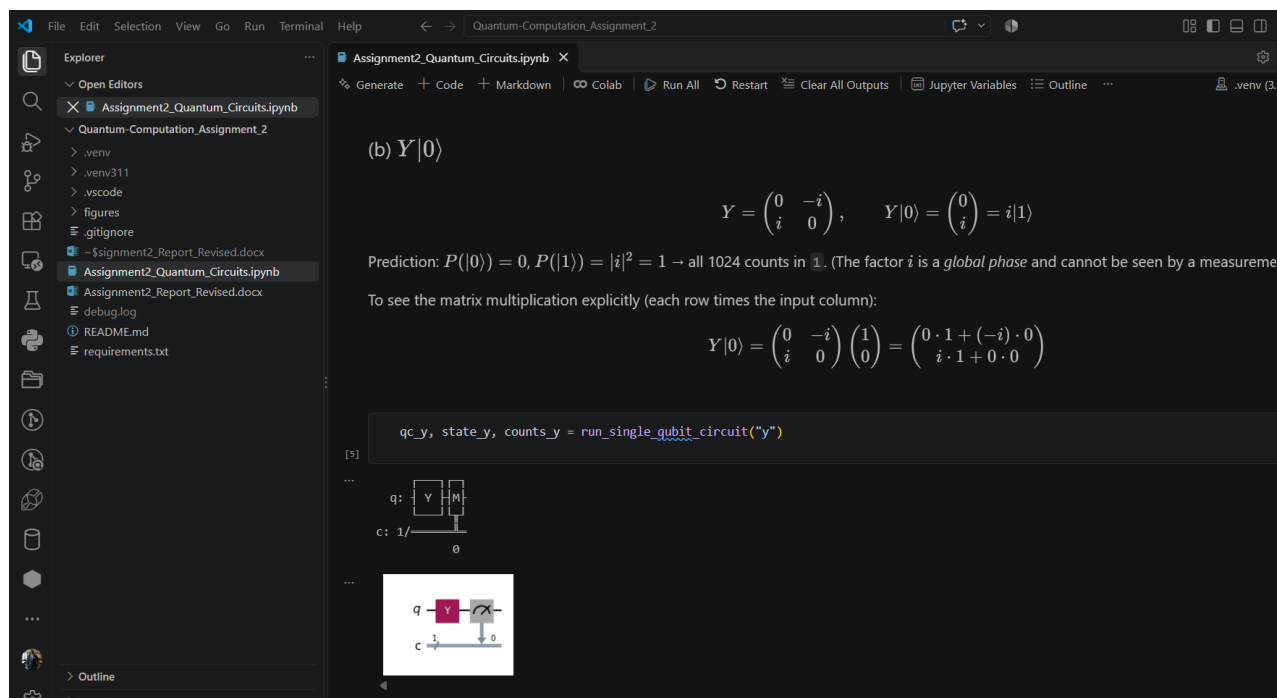
State vector before measurement: `[0.+0.j, 0.+1.j]`



$Y|0\rangle$: measurement histogram and executed-notebook screenshot.

(ii) Circuit screenshot for $Y|0\rangle$ after running draw()

The measured circuit is drawn with `qc.draw("mpl")`. It contains one Y gate on qubit 0 followed by measurement into classical bit 0.



$Y|0\rangle$: circuit diagram and executed-notebook screenshot.

(iii) $Y|0\rangle$ - Pauli Y gate applied to $|0\rangle$
 Derive the output state and probabilities; Verify histogram

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$Y|0\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \times 1 + (-i) \times 0 \\ i \times 1 + 0 \times 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ i \end{bmatrix} = i \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$Y|0\rangle = i|1\rangle$$

Predicted measurement probabilities for $Y|0\rangle$

$$P(|0\rangle) = |0|^2 = 0, P(|1\rangle) = |i|^2 = i^* i = -i^2 = 1.$$

The probabilities sum to 1. The expected counts are 0 zeros and 1024 ones. The state vector agrees with Qiskit's $[0. + 0.j, 0. + 1.j]$. The factor i is a global phase and does not affect measurement probabilities.

- Verification against the $Y|0\rangle$ histogram

The histogram has 0 counts for 0 and 1024 counts for 1. The observed frequencies are $0/1024 = 0$ and $1024/1024 = 1$, exactly matching the mathematical predictions. Every ideal shot produces 1; there is no sampling spread for a probability of 1.

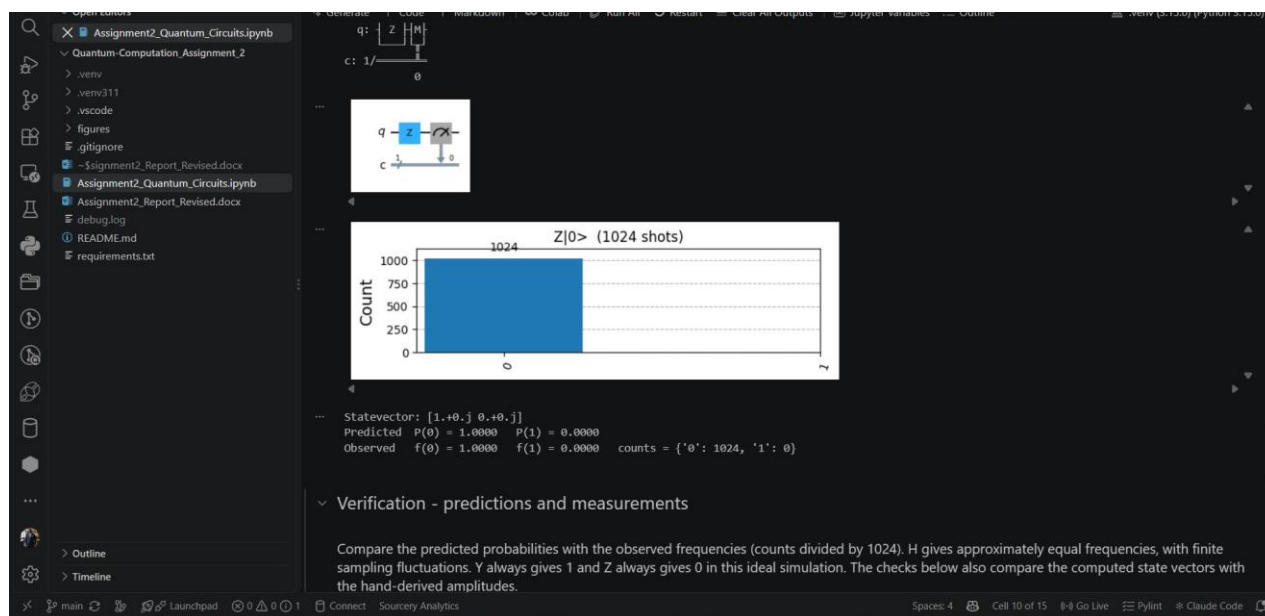
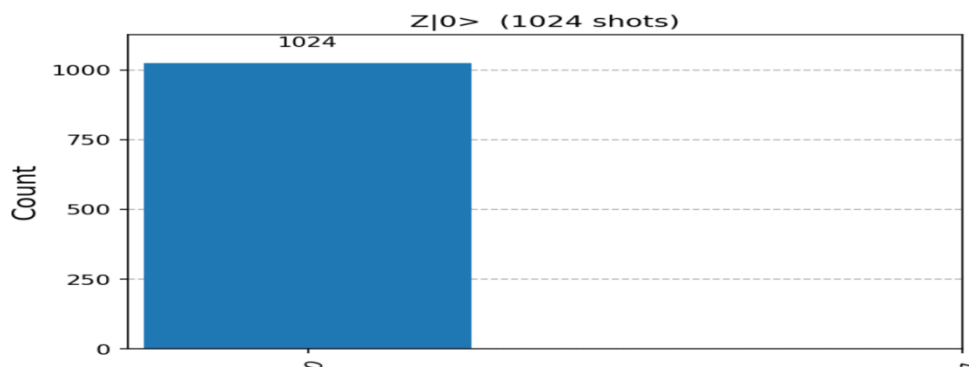
(c) $Z|0\rangle$ — Pauli Z gate applied to $|0\rangle$

(i) Measurement histogram for $Z|0\rangle$

Apply $qc.z(0)$ to `QuantumCircuit(1, 1)`. Run `qc_z, state_z, counts_z = run_circuit("z")` using the shared setup. Measure with `qc.measure(0, 0)`.

1024-shot output: `{'0': 1024, '1': 0}`

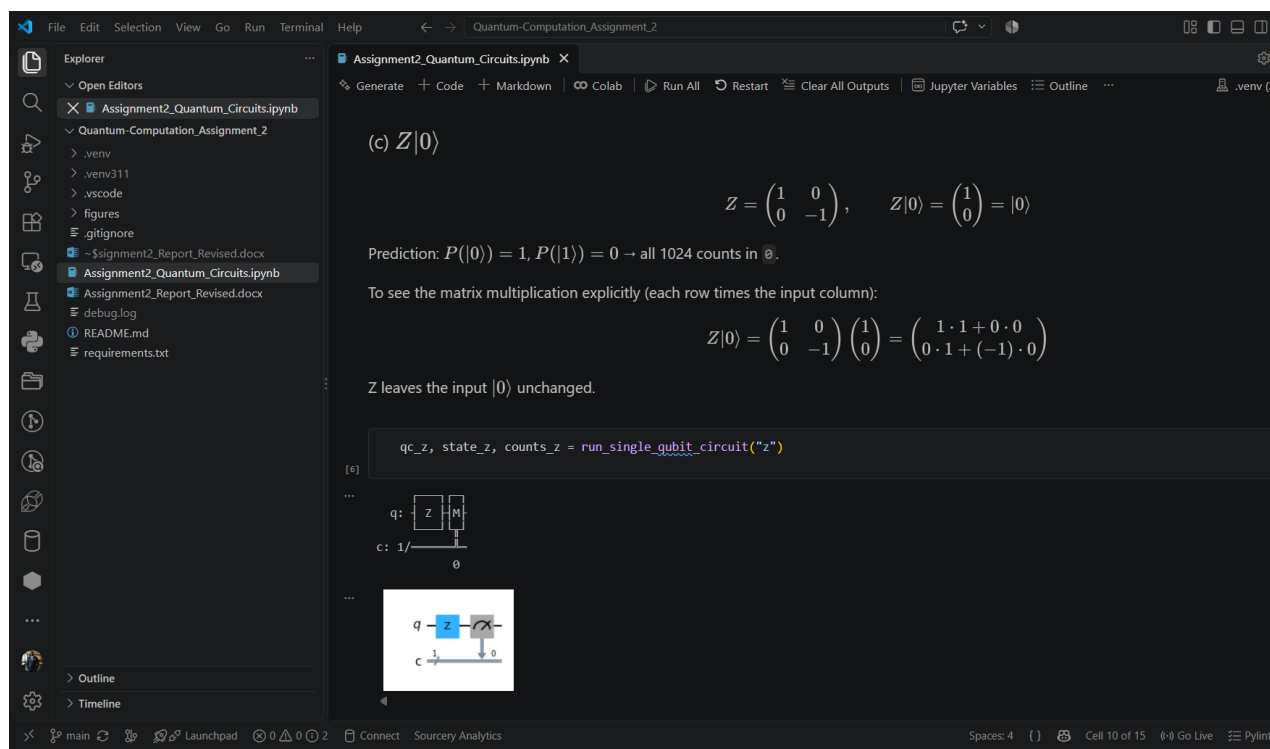
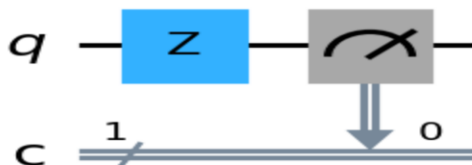
State vector before measurement: `[1.+0.j, 0.+0.j]`



$Z|0\rangle$: measurement histogram and executed-notebook screenshot.

(ii) Circuit screenshot for $Z|0\rangle$ after running draw()

The measured circuit is drawn with `qc.draw("mpl")`. It contains one Z gate on qubit 0 followed by measurement into classical bit 0.



$Z|0\rangle$: circuit diagram and executed-notebook screenshot.

(iii) $Z|0\rangle$ - Pauli Z gate applied to $|0\rangle$
 Derive the output state and probabilities, verify Histogram.

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \times 1 + 0 \times 0 \\ 0 \times 1 + (-1)(0) \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Z|0\rangle = |0\rangle$$

Predicted measurement probabilities for $Z|0\rangle$

$$P(|0\rangle) = |1|^2 = 1, \quad P(|1\rangle) = |0|^2 = 0$$

The probabilities sum to 1. The expected counts are 1024 zeros and 0 ones. The state vector agrees with Qiskit's $[1. + 0.j, 0.j]$.

Verification against the $Z|0\rangle$ histogram

The histogram has 1024 counts for 0 and 0 counts for 1. The observed frequencies are $1024/1024 = 1$ and $0/1024 = 0$, exactly matching the predicted probabilities.

Z flips the sign of a $|1\rangle$ amplitude:

$$Z(a_0|0\rangle + a_1|1\rangle) = a_0|0\rangle - a_1|1\rangle.$$

The requested input $|0\rangle$ has no $|1\rangle$ component, so its state is unchanged. For superpositions, $Z|+\rangle = |-\rangle$; observing the relative phase requires a suitable measurement basis.

(iv) One document containing all answers

This report contains the supplied Question 1 answer and the answers to (a), (b), and (c), including code, histogram and circuit screenshots, mathematical predictions, and verification. The table summarizes the results for 1024 shots per circuit.

Circuit	Predicted output state	Predicted P(0), P(1)	Measured counts '0' / '1'	Observed f(0), f(1)	Agree?
$H 0\rangle$	$\frac{1}{\sqrt{2}}(0\rangle + 1\rangle)$	0.5, 0.5	526 / 498	0.514, 0.486	Yes (within 1σ)
$Y 0\rangle$	$i 1\rangle$	0, 1	0 / 1024	0, 1	Yes (exact)
$Z 0\rangle$	$ 0\rangle$	1, 0	1024 / 0	1, 0	Yes (exact)

H agrees within finite-shot variation; Y and Z agree exactly with the ideal predictions. These are ideal AerSimulator results, not measurements from a physical quantum processor.

Reference

IBM Quantum Learning, Basics of Quantum Information — Single systems.

<https://learning.quantum.ibm.com/course/basics-of-quantum-information/single-systems>